The Importance of Course Redesign

Microlearning Strategies for Enhanced Course Design: Insights from the JCSU CITL Summer Workshop

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Abstract

This article explores innovative course design strategies aimed at enhancing student engagement and learning retention, informed by insights gained during the JCSU CITL Summer Course Design Workshop held from June 2nd to the 4th, 2025. Specifically tailored for STEM faculty, the workshop emphasized integrating microlearning techniques, modular course structures, and establishing instructor presence. This paper discusses key takeaways, practical strategies, and the implications of these approaches for improving educational experiences.

Introduction

As a college professor, I am continually seeking innovative methods to improve course design and enhance student engagement. The recent JCSU CITL Summer Course Design Workshop provided valuable insights into current instructional design tools and techniques that can be effectively integrated into teaching practices. Although my university operates on a semester track, the workshop highlighted microlearning as a powerful asset for improving student learning retention and engagement(Robles et al., 2023). This article shares key takeaways from the workshop, including the importance of adopting a modular course structure, establishing instructor presence, utilizing surveys for assessment, and fostering student discussions. Additionally, I discuss practical strategies such as the use of welcome videos and recorded lectures to create a more interactive and supportive learning environment.

A Hands-On Approach to Learning

During the JCSU CITL Summer Course Design Workshop, participants engaged in hands-on sessions utilizing the Canvas Learning Management System (LMS)(Pujasari & Ruslan, 2021). One enlightening aspect was exploring how artificial intelligence can be integrated into course designs. We also examined the Quality Matters framework, which emphasizes best practices in course design, ensuring courses are not only informative but also engaging (Pujasari & Ruslan, 2021). Furthermore, we developed a STEM Toolkit, utilizing library resources to support our teaching efforts.

Identifying Gaps in Course Design

A significant portion of the workshop focused on identifying gaps in course design within the Canvas platform. Participants were encouraged to adopt an honest perspective, viewing their courses through the eyes of their students. Upon evaluating my course homepage, I recognized it was cluttered and underdeveloped. This realization prompted me to consider the concept of microlearning—breaking down information into smaller, more digestible chunks for better processing (Robles et al., 2023). Consequently, I redesigned my syllabus for CSC 131 – Computers in Society to reflect these microlearning principles, organizing content into smaller sections to enhance clarity and improve student engagement.

Research Questions

The following research questions guided the exploration of course design strategies during the JCSU CITL Summer Course Design Workshop:

- **Does microlearning enhance students' ability to follow directions and digest course materials effectively?**
- **How does frequent communication impact student progress—does it help or hinder their learning experience?**
- **What are some creative strategies to engage students and ensure effective learning from the outset?**
- **In course design, how crucial is the role of the instructor in a modular format?**
- **How can critical thinking be effectively incorporated into assessments?**

These questions aim to investigate the relationship between course design elements and student engagement, retention, and overall success.

Specific Activities Utilized

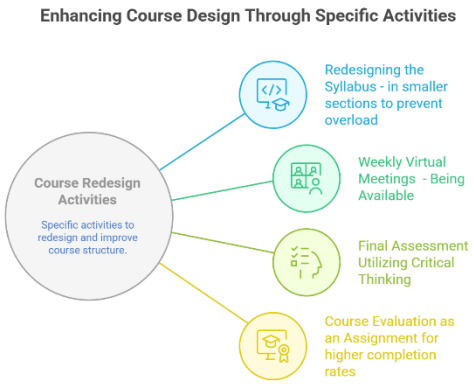
To implement the insights gained from the workshop, several specific activities were utilized in the redesign of the course:

- **Redesigning the Syllabus:** The syllabus was transformed into a slide presentation format, with each slide dedicated to specific information(Ehrlich & Dundis, 2004). For example:
    - One slide detailed grading criteria and rubrics.
    - Another slide contained professor contact information.
    - A separate slide outlined the late policy.

This approach made essential information easily accessible and visually organized, enhancing student understanding.

- **Weekly Virtual Meetings:** To foster human interaction and build rapport, weekly virtual face-to-face meetings were scheduled(Ho et al., 2021). These sessions provided students with opportunities to ask questions, discuss course content, and receive real-time feedback.

- **Final Assessment:** The final assessment was designed as a critical thinking assignment, aimed at evaluating students' understanding of the course material throughout the semester(Singh, 2017). A detailed rubric was provided to guide students on the expectations and criteria for assessment(Barney et al., 2011).
- **Course Evaluation as an Assignment:** Placing the course evaluation as an assignment encouraged students to reflect on their learning experiences and provide constructive feedback(Bukhari et al., 2021). This not only informed course improvements but also enhanced student engagement by making them active participants in the evaluation process(Choudhary, 2024).



Deepening Analysis of Key Takeaways

Evaluation Results

In the Summer 2025 session of CSC 131 – Computers in Society, the course had an enrollment of 12 students, with 10 responding to the evaluation, resulting in a response rate of 83.33%. The feedback gathered provides valuable insights into the effectiveness of the course design and areas where improvements can be made.

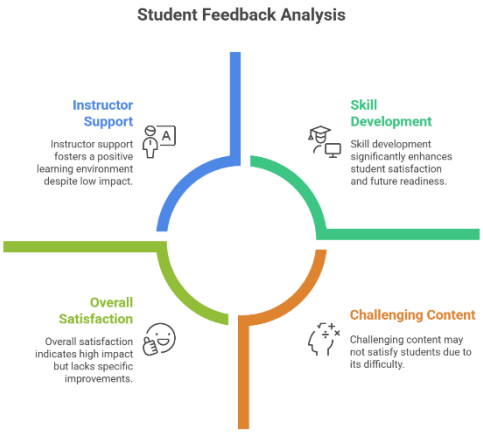
Student Feedback

The responses from students highlighted several strengths of the course, as well as potential areas for enhancement:

1. **Skill Development:**
    - **Response 1:** "I liked CSC 131; it gave me useful tech skills that I can carry on to my college days. The certifications kept me going and gave me a way to measure the progress."
    - **Analysis:** This feedback underscores the effectiveness of the course in imparting practical skills that students find valuable for their future studies. The incorporation of certifications appears to be a motivating factor, providing students with tangible milestones for their learning progress.

2. **Instructor Support:**
    - **Response 2:** "Awesome Professor, always supportive when asked questions."

- **Analysis:** The positive comments regarding instructor presence indicate that establishing a supportive learning environment is crucial for student engagement. This aligns with the workshop's emphasis on instructor presence as a key component of effective course design.



3. **Challenging Content:**
   - **Response 3:** "This course was a good challenge for me to think outside the box and to use my resources more efficiently."
   - **Analysis:** Students appreciate the challenging nature of the course, which encourages critical thinking and resourcefulness. This feedback reinforces the importance of designing courses that stimulate intellectual engagement and foster problem-solving skills.

4. **Overall Satisfaction:**
   - **Response 4:** "None, it was a great course."
   - **Response 5:** "Nothing."
   - **Analysis:** The absence of negative feedback from some students indicates a high level of overall satisfaction with the course. However, it is essential to consider that the lack of specific suggestions for improvement may also reflect a need for more structured feedback mechanisms.

Areas for Improvement

While the feedback was largely positive, there were suggestions for enhancing the course further:

- **Incorporation of More Interactive Elements:**
  - **Response 1** also mentioned that "modules, short quizzes, or practice challenges would make this course even better." This highlights an opportunity to incorporate more interactive and formative assessments, such as quizzes and practice challenges, to reinforce learning and maintain engagement(Clinton-Lisell, 2018).

The evaluation results from the Summer 2025 session of CSC 131 provide valuable insights into the course's effectiveness. The positive feedback regarding skill development, instructor support, and overall satisfaction indicates that the strategies implemented align well with the goals of

enhancing student engagement and learning retention(Hettiarachchilagea & Haldolaarachchige, 2023). However, incorporating more interactive elements could further elevate the course experience. This feedback will guide future iterations of the course design, ensuring continuous improvement and responsiveness to student needs(Cao, 2023).

Expand on Practical Strategies

The incorporation of practical strategies in CSC 131 – Computers in Society has proven effective in enhancing student engagement and learning outcomes. Below are key strategies implemented in the course, along with examples of their successful application:

Modular Course Structure

The course was redesigned to follow a modular structure, allowing students to navigate content in smaller, manageable sections(Vahid & Allen, 2024). Each module included specific objectives, readings, and assessments, which facilitated a clearer understanding of expectations(Álvarez et al., 2023). This approach not only improved organization but also helped students track their progress more effectively.

Use of Assessments and Certifications

Incorporating assessments such as quizzes and certifications served as motivational tools for students(Shankar et al., 2013). The certifications provided a tangible way for students to measure their progress and gain recognition for their achievements(Piontek, 2008). For instance, one student completed the final project with a **100% completion rate**, exemplifying the effectiveness of this strategy in fostering student success.

Here is a clear picture of the student completing their final project, showcasing their dedication and achievement:

Final Project Completion Example

Welcome Videos

The use of welcome videos at the beginning of the course helped establish a personal connection with students, setting a positive tone for the semester(İzmirli & İzmirli, 2019). These informal introductions allowed students to feel more comfortable and engaged from the outset, fostering a welcoming atmosphere conducive to learning.

Interactive Discussions

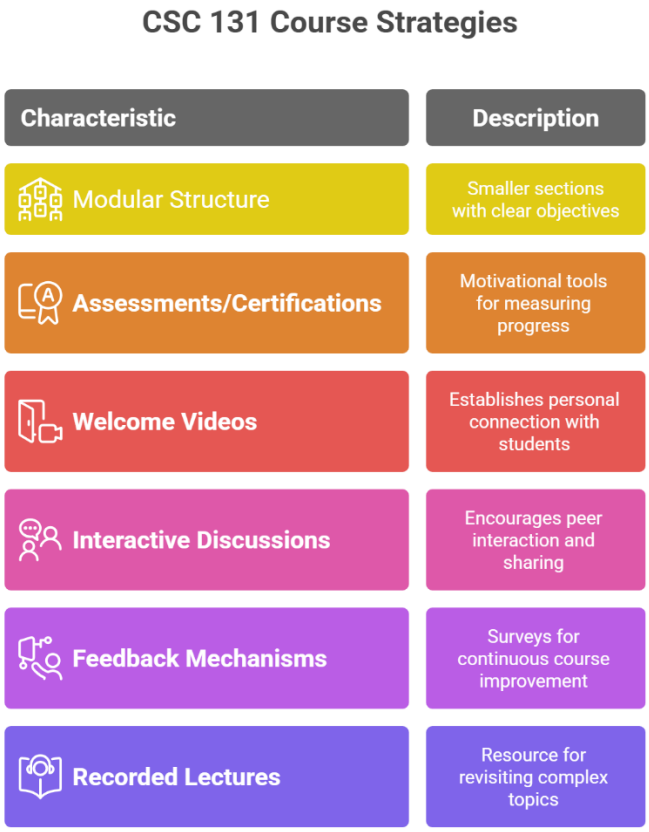
Facilitating discussions among students was another critical strategy(Barker et al., 2014). By posing thought-provoking questions and encouraging peer interaction, I created an environment where students felt empowered to share their insights and experiences. This not only built a sense of community but also enhanced critical thinking skills.

Feedback Mechanisms

Implementing regular feedback mechanisms, such as surveys, allowed for continuous improvement of the course(Syifaulkaromah et al., 2021). Collecting feedback at the beginning and end of the semester provided insights into student perceptions and course effectiveness. This iterative process ensured that the course remained responsive to student needs.

Recording Lectures

Recording lectures served as a valuable resource for students, enabling them to revisit complex topics and reinforce their understanding(Arnold, 2019). These recordings, available on my faculty page, enhanced accessibility and provided additional support for students outside of class hours.



The practical strategies employed in CSC 131 have significantly contributed to student engagement and success. By utilizing a modular structure, assessments, welcome videos, interactive discussions, feedback mechanisms, and recorded lectures, I have created a more dynamic and supportive learning environment(Vahid & Allen, 2024). The example of the student who completed their final project with a 100% completion rate serves as concrete evidence of the effectiveness of these strategies in promoting student achievement.

Discuss Future Implications

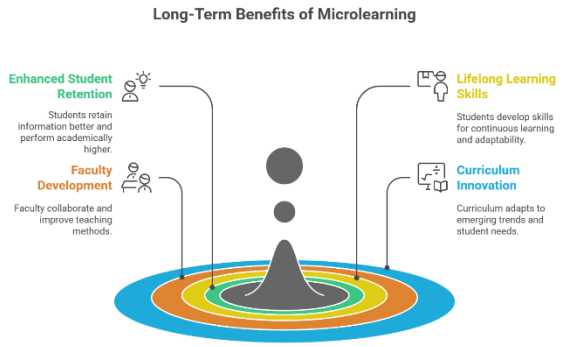
Long-Term Impact

The adoption of microlearning and modular structures in course design has the potential to create significant long-term benefits for both student outcomes and faculty development:

1. **Enhanced Student Retention and Engagement:**
    - Microlearning allows students to engage with content in bite-sized segments, making it easier to retain information(Samala et al., 2023). Over time, this can lead to improved academic performance and a deeper understanding of complex topics.
    - Modular structures promote flexibility, enabling students to learn at their own pace(Major & Calandrino, 2018). This adaptability can lead to higher levels of motivation and satisfaction, ultimately resulting in increased retention rates.
2. **Development of Lifelong Learning Skills:**

- By incorporating microlearning techniques, students can develop essential skills such as self-directed learning and time management(Thongmak, 2021). These skills are crucial for success in both academic and professional settings, fostering a mindset of continuous improvement and adaptability.

3. **Faculty Development and Collaboration:**
    - The shift to modular course design encourages faculty to collaborate and share best practices(Bhardwaj et al., 2025). This can lead to the creation of a supportive community of educators who continuously refine their teaching strategies based on shared experiences and student feedback.
    - Faculty who embrace microlearning may also become more adept at integrating technology into their teaching, enhancing their overall instructional effectiveness(Fidan, 2023).
4. **Curriculum Innovation:**
    - The modular approach allows for easier updates and modifications to course content in response to emerging trends and student needs(Tomei, 2013). This flexibility can lead to a more relevant and timelier curriculum that prepares students for the evolving demands of their fields.



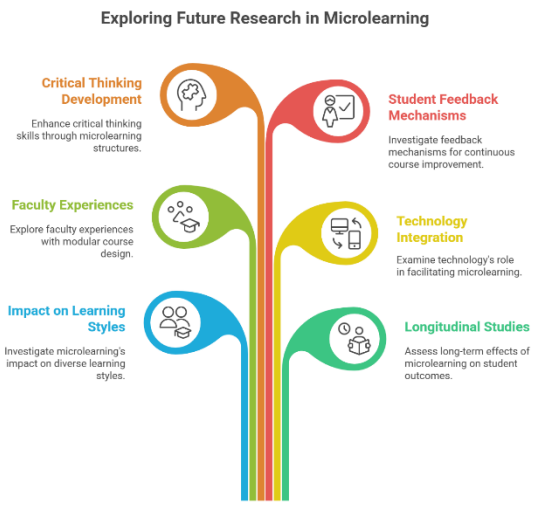
Further Research

To build on the findings from this course design initiative, several areas for future research and exploration could be considered:

1. **Impact of Microlearning on Diverse Learning Styles:**
    - Investigate how microlearning strategies affect students with different learning styles or preferences(Zhu et al., 2024). Understanding these dynamics can help educators tailor their approaches to maximize engagement and effectiveness.
2. **Longitudinal Studies on Student Outcomes:**

- Conduct longitudinal studies to assess the long-term effects of microlearning and modular structures on student performance, retention, and satisfaction across multiple courses and disciplines(Oliveira et al., 2023).

3. **Faculty Experiences with Modular Design:**
    - Explore the experiences of faculty who have transitioned to modular course design. Researching their challenges, successes, and strategies could provide valuable insights for other educators considering similar changes(Mentzer et al., 2022).
4. **Technology Integration in Microlearning:**
    - Examine the role of technology in facilitating microlearning. Investigating the effectiveness of various digital tools and platforms can help identify best practices for implementation(Shaffer et al., 2015).
5. **Critical Thinking Development:**
    - Further research could focus on how microlearning and modular structures can specifically enhance critical thinking skills among students(Rahayu et al., 2022). Identifying effective assessment methods for measuring these skills would also be beneficial.
6. **Student Feedback Mechanisms:**
    - Investigate the effectiveness of different feedback mechanisms on student learning and engagement within a modular framework. Understanding how to best gather and utilize student feedback can lead to continuous improvement in course design(Gedye, 2010).

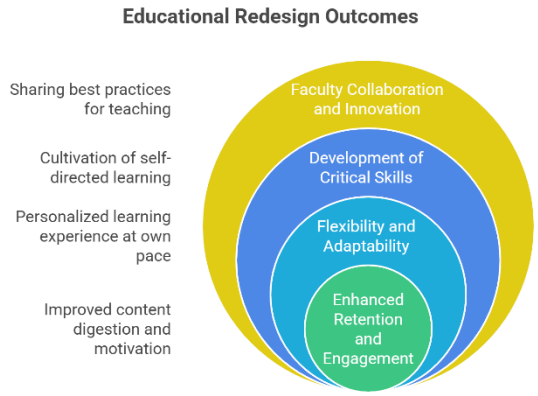


Conclusion

The long-term implications of adopting microlearning and modular structures in course design are profound, with the potential to enhance student learning outcomes and foster faculty development(Kossen & Chia-Yi, 2021). By exploring these areas for future research, educators can continue to refine their approaches and contribute to the ongoing evolution of effective teaching practices(Daff et al., 2024).

The insights gained from the Johnson C. Smith University Center for Innovative Teaching and Learning Summer Course Design Workshop, I was able to redesign CSC 131 – Computers in Society to highlight the significant impact of adopting microlearning and modular structures on student engagement and learning outcomes. Key findings include:

- **Enhanced Retention and Engagement:** Microlearning allows students to digest content in manageable segments, leading to improved retention and motivation(Major & Calandrino, 2018).
- **Flexibility and Adaptability:** Modular course structures empower students to learn at their own pace, fostering a more personalized educational experience(Samala et al., 2023).
- **Development of Critical Skills:** These approaches not only enhance academic performance but also cultivate essential skills such as self-directed learning and critical thinking(Bhardwaj et al., 2025).
- **Faculty Collaboration and Innovation:** The transition to modular design encourages faculty collaboration and the sharing of best practices, ultimately leading to continuous improvement in teaching strategies(Sakhieva et al., 2014).



As we move forward in the evolving landscape of education, it is crucial for educators to consider the implementation of microlearning and modular structures in their courses(Short & Hirsh, 2020). By adopting these innovative strategies, educators can create more engaging and effective learning environments that cater to the diverse needs of their students(Robles et al., 2023).

I encourage fellow educators to experiment with these approaches in their own teaching practices and to share their experiences and insights. By fostering a community of practice, we can collectively enhance the quality of education and better prepare our students for the challenges of the future(Kalantzis & Cope, 2004). Together, let’s embrace the potential of microlearning and modular design to transform teaching and learning in meaningful ways.

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